Check for updates

RESEARCH ARTICLE doi.org/10.1002/adsc.70091

ADVANCED SYNTHESIS & CATALYSIS

Organocatalytic Remote-Controlled Enantioselective 1,10-Additions of In Situ Generated Alkynyl 2,4′-Biphenyl Quinone Methides

Tian Liang,[a] Meiwen Liu,[b] Aman Ullah,[c] Zhibin Yue,[a] Yan Xia,[a] Fang Fang,[b] Pengfei Li,[b,*] Shao-Fei Ni,[c,*] and Wenjun Li[a,*]

[a] Department of Medicinal Chemistry, School of Pharmacy, Qingdao University, Qingdao, Shandong 266021, China
E-mail: liwj@qdu.edu.cn
[b] Department of Chemistry, Guangdong Provincial Key Laboratory of Catalysis, College of Science, Southern University of Science and Technology Guangming Advanced Research Institute, Southern University of Science and Technology (SUSTech), Shenzhen 518055, China
E-mail: lipf@sustech.edu.cn
[c] College of Chemistry & Chemical Engineering, Chemistry and Chemical Engineering Guangdong Laboratory, Shantou University, Shantou 515063, China
E-mail: sfni@stu.edu.cn

Manuscript received: June 24, 2025; Revised manuscript received: August 11, 2025;
Version of record online: September 30, 2025

Supporting information for this article is available on the WWW under https://doi.org/10.1002/adsc.70091

Abstract: The strategic incorporation of auxiliary groups has recently established α-functionalized alcohols as versatile electrophilic alkylation reagents, enabling remotely controlled conjugate additions. Strikingly, while quinone methide chemistry has been extensively developed, their extended analogues—featuring spatially separated carbonyl and methide units across biphenyl systems—remain unexplored. Building on the prior work with alkynyl 4,4′-biphenyl quinone methides, an organocatalytic remotely controlled enantioselective 1,10-additions of in situ-generated alkynyl 2,4′-biphenyl quinone methides from α-[4-(2- hydroxyphenyl)phenyl]propargyl alcohols with 3-arylindoles and indole-2-carboxylates is reported, affording a broad scope of enantioenriched 3*H*-pyrrolo[1,2-*a*]indoles and axially chiral tetrasubstituted allenes, respectively. Combined with control experiments, density functional theory calculations elucidate the reaction mechanism, revealing that the dehydration of α-functionalized alcohols to generate 2,4′-biphenyl quinone methides constitutes the rate-limiting step, while enantioselectivity arises from the nucleophilic addition of alkynyl biphenyl quinone methides. Notably, beyond proving 2,4′-biphenyl quinone methides as competent intermediates, this work breaks new ground by demonstrating their participation in previously unreported stereoselective 1,10-additions.

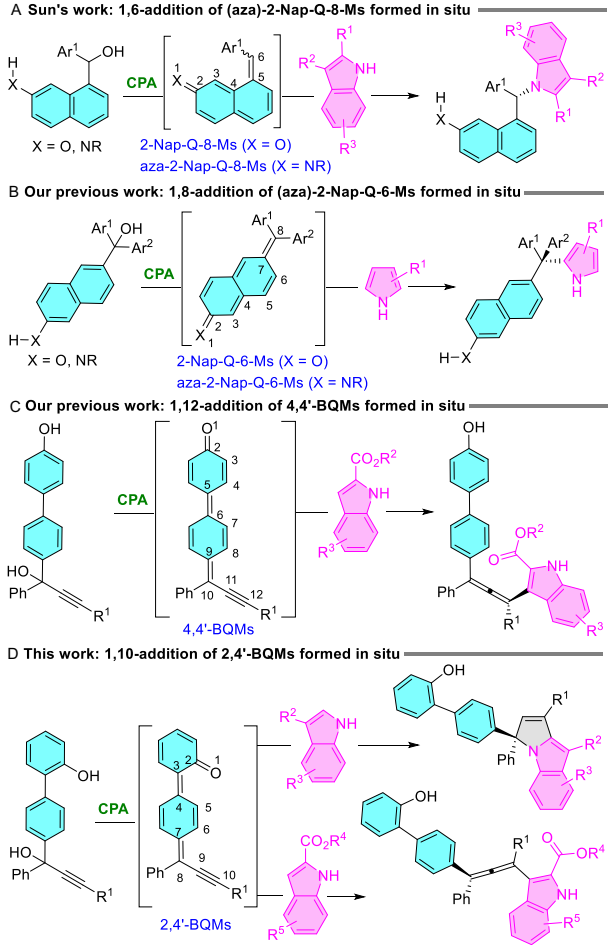
Keywords: 1,10-additions, 3-arylindoles, indole-2-carboxylates, organocatalysis, quinone methides

1. Introduction

The strategic design of auxiliary groups has revolutionized α-functionalized alcohol chemistry in recent years, establishing these substrates as versatile platforms for stereoselective electrophilic alkylation and related transformations.[1] In this context, phenol and aniline have emerged as particularly practical auxiliary groups, enabling their corresponding α-functionalized alcohols to in situ generate high-value quinone methides (QMs) and aza-QMs, respectively.[2] These reactive intermediates serve as versatile platforms for diversity-oriented synthesis.[3] In striking contrast to well-developed QM chemistry, organocatalytic stereoselective reactions of their extended analogues—where carbonyl and methide units reside on separate aromatic rings—remain exceptionally rare. This stems from the inherent challenge of exerting stereocontrol over spatially remote reactive centers in these systems.[4] To date, only three landmark examples have been reported in this emerging field. In 2023, Sun and coworkers achieved a breakthrough with the first CPA-catalyzed asymmetric 1,6-conjugate addition to in situ-generated 2-naphthoquinone-8-methides (2-Nap-Q-8-Ms) and their aza-analogues (aza-2-Nap-Q-8-Ms), derived from 8-(hydroxyarylmethyl)naphthalen-2-ols/amines for the synthesis of previously less accessible remotely chiral naphthols and naphthylamines

 © 2025 Wiley-VCH GmbH

(**Scheme 1**A).[5] Simultaneously, we developed the complementary CPA-catalyzed 1,8-addition to (aza)-2-naphthoquinone-6-methides ((aza)-2-Nap-Q-6-Ms) from 6-(hydroxydiarylmethy l)naphthalen-2-ols/amines (Scheme 1B),[6] enabling efficient synthesis of chiral tetraarylmethanes.[7] Continually, we further pioneered stereoselective 1,12-additions using alkynyl 4,4′-biphenyl quinone methides (4,4′-BQMs) generated from *α*-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols for construction of axially chiral tetrasubstituted allenes (Scheme 1C).[8] Collectively, these advances systematically expand the auxiliary group design from basic phenol/phenylamine to extend naphthol/naphthylamine and biphenyl systems, creating a versatile platform for CPA-catalyzed remote functionalization of *α*-substituted alcohols. To further explore the chemistry of BQMs,[9] and as our going interesting in developing *α*-functionalized alcohols,[6,8,10] we sought to expand the system from alkynyl 4,4′-BQMs to their 2,4′-BQM counterparts, focusing on the regiochemical flexibility of QM formation, the impact of altered *π*-conjugation patterns on reactivity, and new opportunities for stereocontrol in subsequent transformations. Herein, we report a catalytic asymmetric 1,10-addition to in situ-generated alkynyl 2,4′-BQMs derived from *α*-[4- (2-hydroxyphenyl)phenyl] propargyl alcohols (Scheme 1D).[11] This remotely controlled transformation demonstrates remarkable substrate-directed divergence: 1) With 3-arylindoles, the reaction constructs enantioenriched 3*H*-pyrrolo[1,2-*a*]indoles via a dehydration/conjugate addition/annulation cascade; 2) Using indole-2-carboxylates, it forges axially chiral tetrasubstituted allenes through a stereoselective dehydration/conjugate addition sequence. Notably, this work not only establishes alkynyl 2,4′-biphenyl quinone methides (BQMs) as versatile intermediates for organocatalytic asymmetric transformations but also demonstrates their unique capacity to enable remote stereocontrolled 1,10-conjugate additions and divergent synthesis of enantioenriched 3*H*-pyrrolo[1,2-*a*]indoles[10,12] and axially chiral tetrasubstituted allenes.[13]



Scheme 1. Reactions of *α*-functionalized alcohols.

2. Results and Discussion

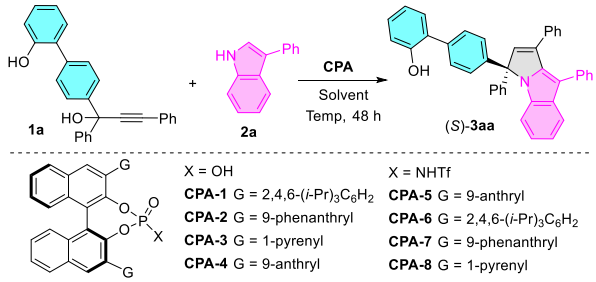
We started our investigation with the model reaction of *α*-[4-(2-hydroxyphenyl)phenyl]propargyl alcohol 1a and 3-phenylindole 2a in dichloromethane at room temperature for 48 h. Preliminary study employing chiral phosphoric acid CPA-1 as catalyst confirmed the formation of product (*S*)-3aa via dehydration-conjugate addition-annulation cascade reaction, albeit with poor enantioselectivity (12% ee, **Table 1**, entry 1).[14] Following systematic evaluation of chiral phosphoric acid catalysts (Table 1, entries 2–8), CPA-2 emerged as the optimal catalyst, delivering product (*S*)-3aa in 96% yield with 88% ee (Table 1, entry 2). To further enhance enantioselectivity, we investigated solvent effects (Table 1, entries 9–15). Pleasingly, employing $CHCl_3$ as reaction medium in the CPA-2 catalyzed transformation afforded (*S*)-3aa in 97% yield with 90% ee (Table 1, entry 9). However, lowering the reaction temperature significantly reduced the product yield (Table 1, entry 16), while variations in concentration showed similar results (Table 1, entries 17–18).

With the optimal conditions in hand, we then examined the substrate scope of the **CPA-2** catalyzed reaction of propargyl alcohols **1** and 3-arylindoles **2** (**Scheme 2**). Firstly, the scope of propargyl alcohols **1** was investigated by the **CPA-2** catalyzed reactions of 3-phenylindole **2a**. Generally, all probed propargyl alcohols **1a-j** reacted smoothly with 3-arylindole **2** to furnish the desired products (*S*)-**3aa-ja** in 91–99% yields with 85–90% ee. Various substituents (R^1) bearing either electron-withdrawing (F, Cl, Br) or electron-donating groups (Me, MeO) were compatible with a slight effect on reaction efficiency and asymmetric induction. Neither significant electronic nor steric effect of the aromatic moiety was observed. Particularly, the reactions of propargyl alcohols with sterically hindered residue ($R^1 = 2\text{-}ClC_6H_4$, $2\text{-}MeC_6H_4$) also proceeded smoothly

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Conditions optimization of reaction between **1a** and **2a**.



Entry[a]	CPA	Solvent	Yield [%][b]	ee [%][c]
1	**CPA-1**	CH_2Cl_2	63	12
2	**CPA-2**	CH_2Cl_2	96	88
3	**CPA-3**	CH_2Cl_2	85	85
4	**CPA-4**	CH_2Cl_2	83	87
5	**CPA-5**	CH_2Cl_2	76	9
6	**CPA-6**	CH_2Cl_2	63	12
7	**CPA-7**	CH_2Cl_2	96	88
8	**CPA-8**	CH_2Cl_2	85	85
9	**CPA-2**	$CHCl_3$	83	87
10	**CPA-2**	CCl_4	76	9
11	**CPA-2**	$ClCH_2CH_2Cl$	63	12
12	**CPA-2**	PhCl	96	88
13	**CPA-2**	$PhCF_3$	85	85
14	**CPA-2**	toluene	83	87
15	**CPA-2**	THF	76	9
16[d]	**CPA-2**	$CHCl_3$	63	12
17	**CPA-2**	$CHCl_3$ (0.6 mL)	96	88
18	**CPA-2**	$CHCl_3$ (1.0 mL)	85	85

[a] Unless noted, a mixture of **1a** (0.06 mmol), **2a** (0.05 mmol), and **CPA** (2 mol%) in the solvent (0.3 mL) was stirred at 25 °C for 48 h;
[b] Isolated yield;
[c] Determined by chiral-HPLC analysis;
[d] At 0 °C for 48 h.

to afford the corresponding products (*S*)-**3ka** in 90% yield with 80% ee and (*S*)-**3la** in 98% yield with 76% ee, respectively. Furthermore, the heteroaryl substituted propargyl alcohol **1m** was also tolerated to generate the desired product (*S*)-**3ma** in 98% yield with 90% ee. Importantly, the **CPA-2** catalyzed reaction could be scaled up to 1.25 mmol, affording the product (*S*)-**3aa** in 92% yield without compromising the enantioselectivity. The absolute configuration of product (*S*)-**3aa** was confirmed by X-ray crystallography (CCDC 2440306). The absolute configurations of other products **3** were assigned by analogy.

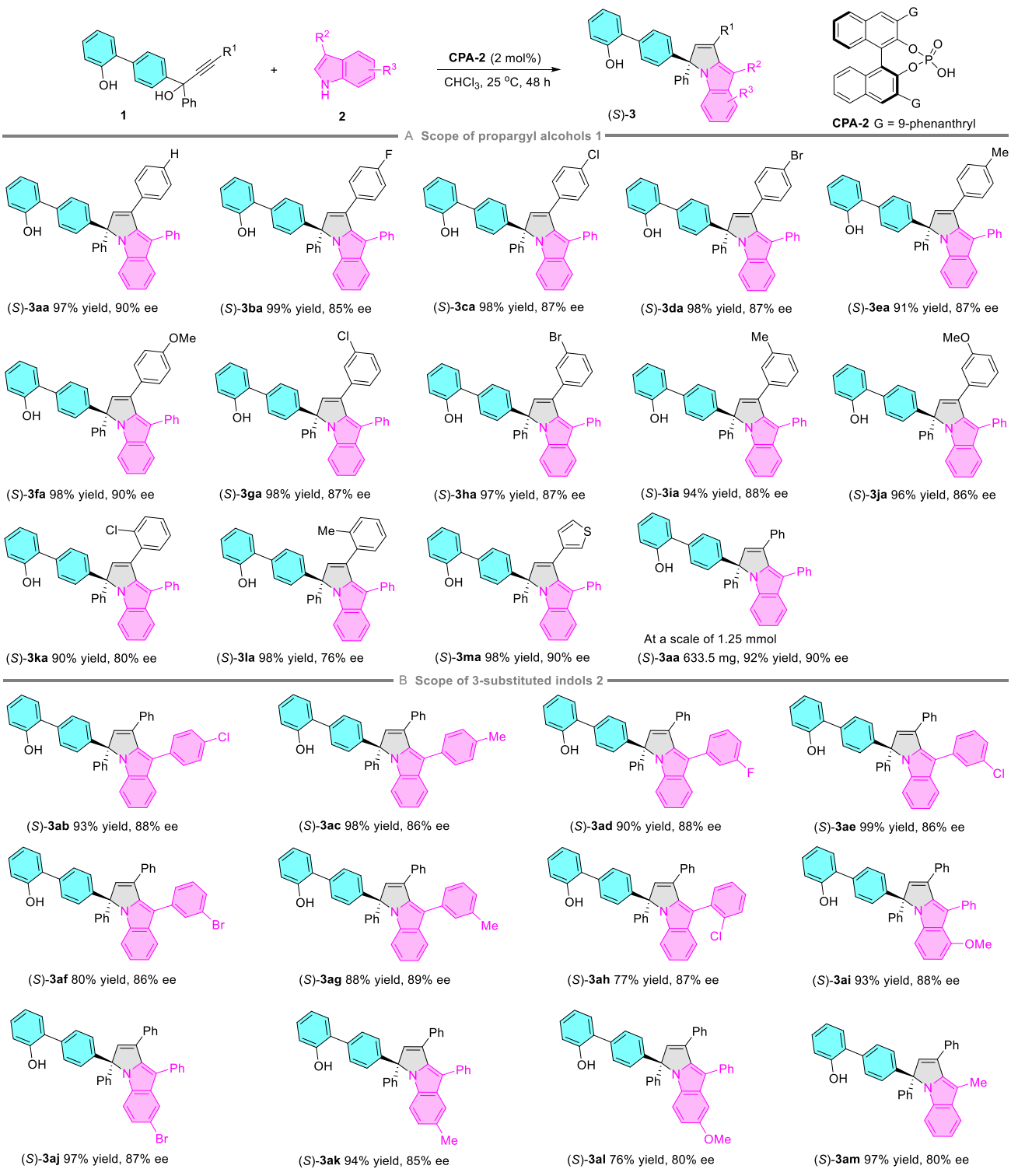
With these encouraging data in hand, we subsequently investigated the scope of 3-arylindoles **2** by the **CPA-2** catalyzed reaction of propargyl alcohol **1a**. Notably, a broad scope of 3-arylindoles **2b-l** with different substituents (R^2/R^3) reacted smoothly with propargyl alcohol **1a** to afford the products (*S*)-**3ab-al** in high to excellent yields (77–99%) with high enantioselectivities (80–89% ee). Either electron-withdrawing (F, Cl, Br) or electron-donating groups (Me, MeO) could be introduced into the aromatic ring of 3-arylindoles with a little influence on the yield and ee. Notably, 3-methylindole **2m** was also compatible to afford the corresponding product (*S*)-**3am** in 97% yield with 80% ee. In all, we developed a class of *α*-[4-(2-hydroxyphenyl)phenyl] propargyl alcohols to react smoothly with 3-arylindoles via dehydration-conjugate addition-annulation cascade reaction, affording a series of 3*H*-pyrrolo[1,2-*a*]indoles in high to excellent yields with high enantioselectivities.

Next, we turned our attention to the chiral phosphoric acid catalyzed asymmetric reactions of *α*-[4-(2-hydroxyphenyl)phenyl]propargyl alcohols **1** with indole-2-carboxylates **4** (**Scheme 3**). Initial result demonstrated that chiral phosphoric acid **CPA-2** failed to promote the desired reaction between propargyl alcohol **1a** and indole-2-carboxylate **4a** in CH_2Cl_2. Encouragingly, axially chiral tetrasubstituted allene (*S*)-**5aa** could be obtained in 97% yield with 86% ee in the presence of **CPA-7** in $CHCl_3$ at 25 °C for 24 h after systematic screening of chiral phosphoric acids and optimization of reaction parameters including solvent, additives, concentration, temperature, and time (see Supporting Information for more details).

Subsequently, we examined the scope of the **CPA-7** catalyzed reaction of alcohols **1** and indole-2-carboxylates **4** (**Scheme 4**). Notably, a broad scope of indole-2-carboxylates bearing R^4/R^5 substituents was compatible to afford the desired axially chiral allenes in generally good to high yields and stereoselectivities (Scheme 4A). More specifically, the ester groups of indole-2-carboxylates (CO_2R^4) had a slight effect on the yield and enantioselectivity, affording the corresponding allenes (*S*)-**5aa-af** in 80–97% yields with 70–86% ee, respectively. Regardless of the substituent's electronic or steric properties, indole-2-carboxylates **4g-k** bearing diverse substituents (R^5 = 6-Cl, 6-Br, 5-F, 5-Br, 5-Me) also reacted smoothly with propargyl alcohol **1a** to furnish the desired products (*S*)-**5ag-ak** in 78–97% yields with 75–93% ee. Exceptionally, racemic product (*S*)-**5al** was obtained in 17% yield when 2-phenylindole was employed, suggesting that the substituent at the 2-position significantly influences enantioselectivity. The scope of alcohols **1** was then surveyed by the **CPA-7** catalyzed reaction of indole-2-carboxylate **4a** (Scheme 4B). Pleasingly, a series of propargyl alcohols bearing aromatic group with different halogen (F, Cl, Br) were well-tolerated to afford the corresponding axially chiral allenes **5** in 76–95% yields with 77–89% ee. The reaction demonstrated remarkable insensitivity to both electronic and steric variations of the substituents. Moreover, the sterically hindered propargyl alcohol **1p** (R^1 = 2-naphthyl) was good substrate to afford the product (*S*)-**5pa** in 92% yields with 87% ee. Exceptionally, the aliphatic substituted propargyl alcohol **1q** (R^1 = cyclopropanyl) was not compatible. Importantly, the **CPA-7** catalyzed reaction of

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

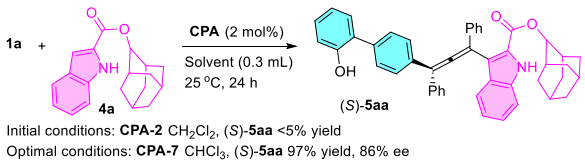


Scheme 2. Substrate scope of reaction between propargyl alcohols **1** and 3-substituted indoles **2**. A mixture of **1** (0.06 mmol), **2** (0.05 mmol), and **CPA-2** (2 mol%) in $CHCl_3$ (0.3 mL) was stirred at 25 °C for 48 h. Product (*S*)-**3** was obtained in isolated yield. The ee was determined by a chiral-HPLC analysis.

alcohol **1a** with indole-2-carboxylate **4a** could be scaled up to 1.25 mmol to afford the target product (*S*)-**5aa** without compromising reaction efficiency and enantioselectivity (96% yield, 86% ee). The absolute configuration of product **5aa** was assigned by comparison between its experimental and calculated electronic circular dichroism (ECD) spectra to be the *S* configuration (**Figure 1**).

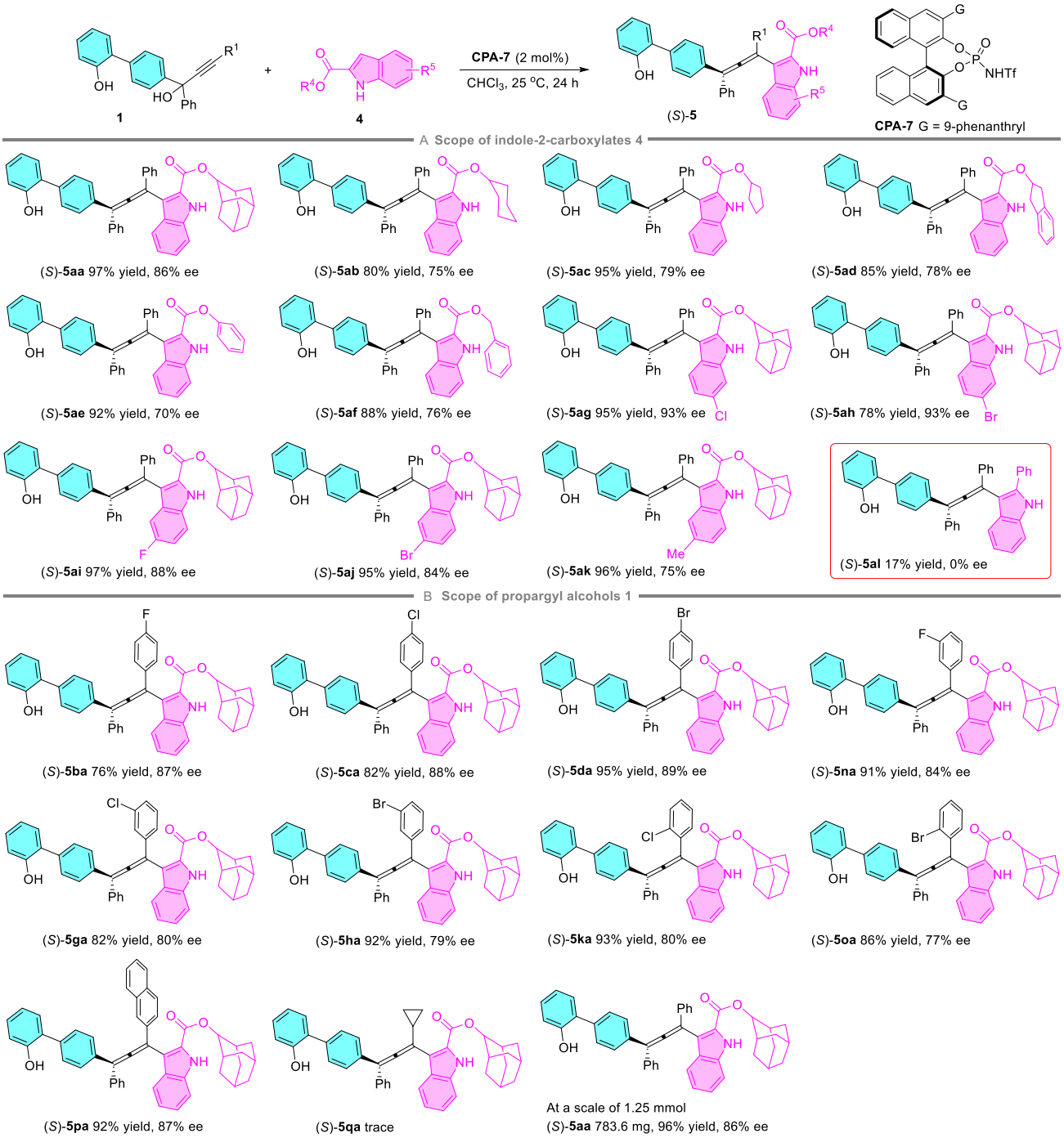
© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 3. Conditions optimization of reaction between **1a** and **4a**.

Taken together, the chiral phosphoric acid catalyzed 1,10-conjugate addition of in situ formed alkynyl 2,4′-biphenyl quinone methides from α-[4-(2-hydroxyphenyl)phenyl]propargyl alcohols with indole-2-carboxylates, furnishing a wide range of axially chiral tetrasubstituted allenes in good to high yields and enantioselectivities.



Scheme 4. Substrate scope of reaction between propargyl alcohols **1** and indole-2-carboxylates **4**. A mixture of **1** (0.05 mmol), **4** (0.10 mmol), and **CPA-7** (2 mol%) in $CHCl_3$ (0.3 mL) was stirred at 25 °C for 24 h. Product (*S*)-**5** was obtained in isolated yield. The ee was determined by a chiral-HPLC analysis.

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

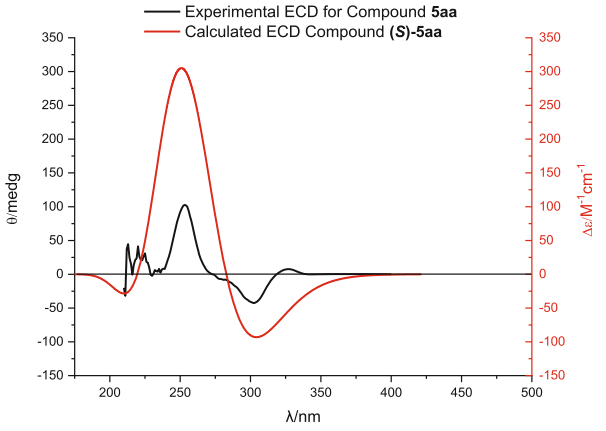
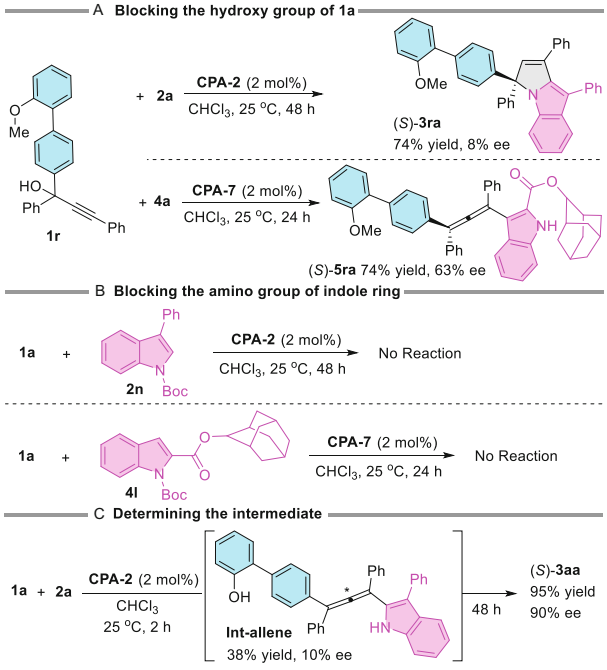


Figure 1. Comparison of the calculated ECD of compound (*S*)-**5aa** with the experimental one of compound **5aa**.

To get insight into the reaction mechanism, further investigations were carried out. As shown in **Scheme 5**A, protection of the hydroxy group in the *α*-[4-(2- hydroxyphenyl)phenyl]propargyl alcohol **1a** with a methyl group resulted in dramatically reduced enantioselectivity, affording products (*S*)-**3ra** (74% yield, 8% ee) and (*S*)-**5ra** (74% yield, 63% ee), respectively. This pronounced erosion of stereocontrol highlights the critical role of the free hydroxyl group in asymmetric induction. Notably, no reaction progression was observed under standard conditions when the nucleophile's amino group was Boc-protected. This complete lack of reactivity, in stark contrast to the free amine system, unambiguously



Scheme 5. Further investigations.

establishes the indispensable role of the free amino group in both reaction initiation and stereochemical control (Scheme 5B). Control experiment revealed the allene intermediate (**Int-allene**) could be isolated in 38% yield with 10% ee from the **CPA-2** catalyzed reaction of **1a** with **2a** after 2 h (Scheme 5C). Notably, this intermediate underwent complete conversion to the desired product (*S*)-**3aa** under standard conditions, achieving 95% yield and 90% ee. These findings demonstrate that the final product's optical purity is established during the **CPA-2**-catalyzed intramolecular annulation of **Int-allene**, rather than during its initial formation.

Density functional theory (DFT) calculations elucidate the **CPA-2**-catalyzed reaction of **1a** with **2a** (**Figure 2**A), revealing a catalytic cycle initiated by hydrogen–bonded complex **2-Int1**. Dehydration via **2-TS1** ($\Delta G^{\ddagger} = 26.3$ kcal mol^{-1}) forms alkynyl 2,4′-BQM **2-Int2**, then forming a hydrogen–bonded complex **2-Int3** with **2a**. Nucleophilic addition of **2a** through **2-TS2** yields **2-Int4**, which undergoes deprotonation via **2-TS3** to give the thermodynamically stable intermediate **2-Int5**. Proton transfer to the allene moiety (**2-TS4**) forms intermediate **2-Int6** ($\Delta G = -18.6$ kcal mol^{-1}), followed by annulation to generate product (*S*)-**3aa**. The rate-determining step involves the formation of the 2,4′-BQM intermediate. The enantioselectivity-determining step involves the conversion of **2-Int6** through transition state **2-TS5-S** ($\Delta G^{\ddagger} = 16.8$ kcal mol^{-1}) to furnish the (*S*)-configured product (*S*)-**3aa** while regenerating the **CPA-2** catalyst. Comparative analysis of the competing transition states revealed a 4.8 kcal mol^{-1} energy difference between the favored **2-TS5-S** and its diastereomeric counterpart **2-TS5-R**, which correlates well with the experimentally observed 90% enantiomeric excess for the (*S*)-product (Figure 2B). To quantitatively elucidate the origin of the 4.8 kcal mol^{-1} energy difference between **2-TS5-R** and **2-TS5-S**, we conducted an energy decomposition analysis in the gas phase ($\Delta\Delta E = 7.2$ kcal mol^{-1}). The results indicate that this energy difference primarily arises from differential distortion energies of both the substrate ($\Delta\Delta E_{sub}$) and the catalyst ($\Delta\Delta E_{cat}$). In **2-Int6**, the substrate adopts a dihedral angle ($\angle 1234 = 41.8°$), which undergoes a more significant change in **2-TS5-R** ($\angle 1234 = 20.9°$) than in **2-TS5-S** ($\angle 1234 = 34.3°$) to facilitate N─C bond formation. Similarly, the catalyst distortion results from a dihedral angle change from $\angle 5,6,7,8 = -119.3°$ in **2-Int6** to $-118.3°$ in **2-TS5-S** and $-68.2°$ in **2-TS5-R**. The substantial steric repulsion in **2-TS5-R** leads to pronounced structural distortion in both the substrate and catalyst, which exhibits significantly higher distortion energy than **2-TS5-S**, indicating that the (*R*)-pathway requires more extensive structural deformation of both components. These findings strongly support that the observed enantioselectivity stems from the more favorable geometric accommodation of the substrate-catalyst system in the (*S*)-configured transition state. Similarly, as shown in **Figure 3**A,

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

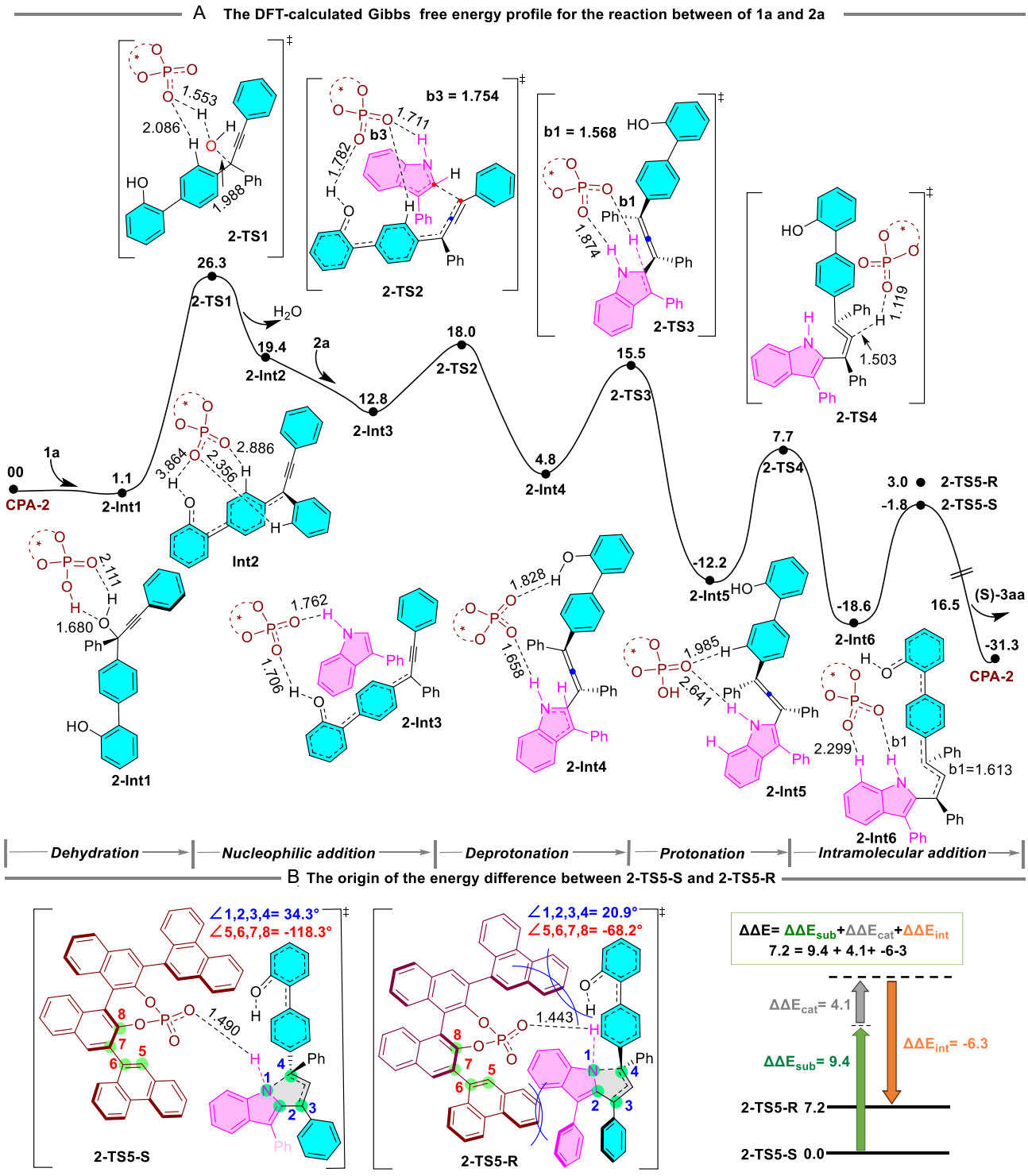


Figure 2. DFT study. A) Gibbs free energy profile for **CPA-2** catalyzed reaction of **1a** with **2a** (the energy values are in kcal mol^{-1}). B) Energy decomposition analysis on the key enantioselectivity-determining transition states.

DFT calculations elucidate the **CPA-7**-catalyzed reaction of **1a** and **4a**, revealing a catalytic cycle initiated by hydrogen–bonded complex **7-Int1** ($\Delta G = 2.1$ kcal mol^{-1}) that undergoes dehydration via **7-TS1** ($\Delta G^{\ddagger} =$ 22.3 kcal mol^{-1}) to form alkynyl 2,4′-BQM **7-Int2**. The enantioselectivity-determining nucleophilic addition occurs through **7-TS2-S** ($\Delta G^{\ddagger} = 18.3$ kcal mol^{-1}), favored by 2.9 kcal mol^{-1} over **7-TS2-R** and consistent with the experimental 86% ee for (*S*)-**5aa**, followed by rearomatization via **7-TS3-S** ($\Delta G^{\ddagger} = 4.6$ kcal mol^{-1}).

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

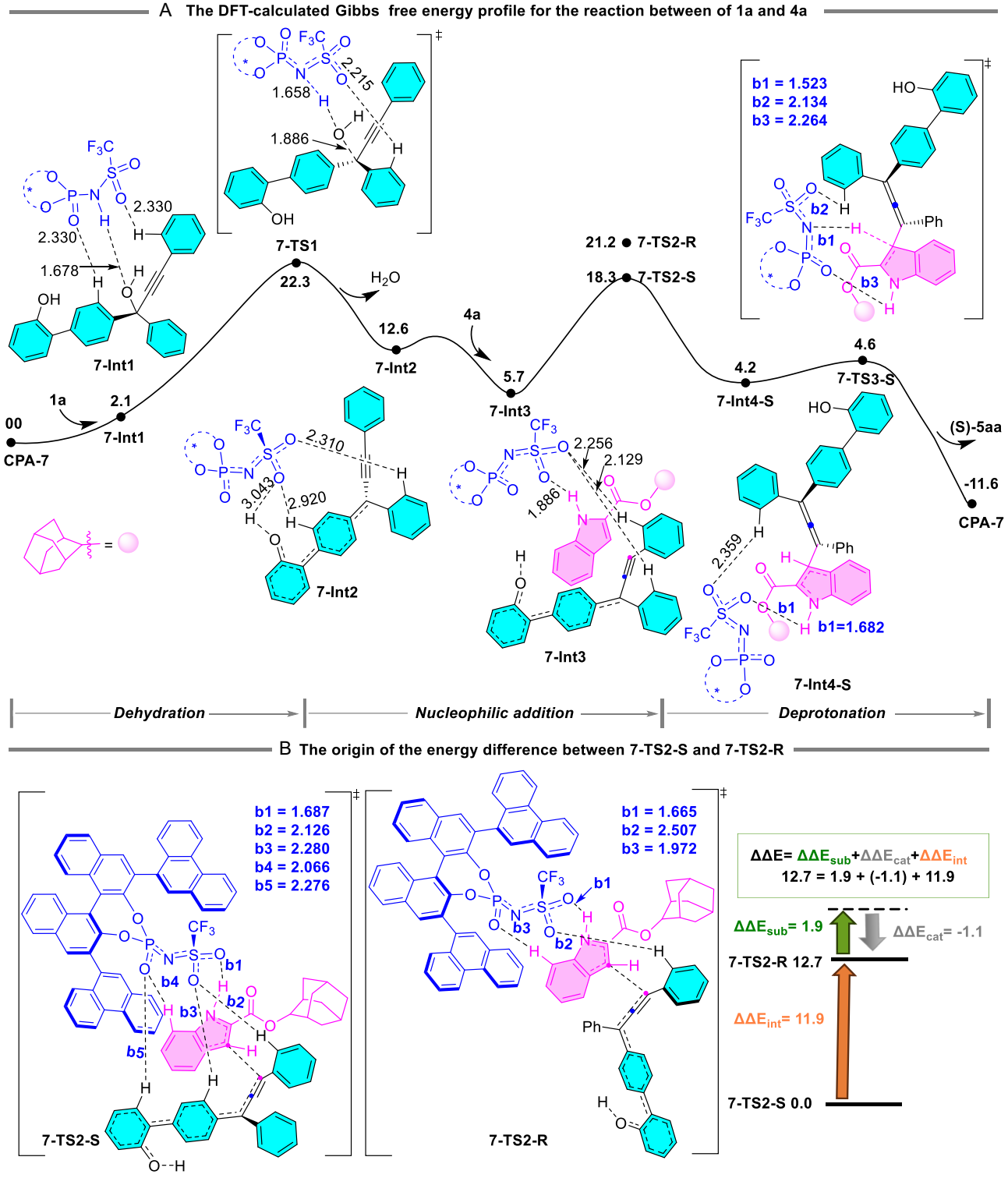


Figure 3. DFT study. A) Free energy profile for the **CPA-7** catalyzed reaction of **1a** with **4a** (the energy values are in kcal mol^{-1}). B) Energy decomposition analysis on the key enantioselectivity-determining transition states.

In Figure 3B, energy decomposition analysis attributes this preference to stronger substrate-catalyst interactions in **7-TS2-S**, which features an optimized network of five hydrogen bonds (S═O···H: 1.687/2.126/2.280 Å; P═O···H: 2.066/2.276 Å) versus only three weaker interactions in **TS2-R** (S═O···H: 1.665/2.507 Å; P═O···H: 1.972 Å), demonstrating how precise non–covalent interaction patterning governs asymmetric induction.

© 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. Conclusions

In conclusion, building on our previous success with 4-phenylphenol-based α-functionalized alcohols, we have now demonstrated that 2-phenylphenol serves as an effective auxiliary group for stereodivergent transformations of the corresponding α-functionalized alcohol. In the presence of CPA, alkynyl 2,4′-BQMs were generated in situ from α-[4-(2-hydroxyphenyl)phenyl] propargyl alcohols and subsequently underwent substrate-controlled enantioselective 1,10-conjugate additions, delivering either enantioenriched 3*H*-pyrrolo[1,2-a]indoles when employing 3-arylindoles as nucleophiles, or axially chiral tetrasubstituted allenes when using indole-2-carboxylates. Comprehensive mechanistic studies combining control experiments and DFT calculations not only confirmed the intermediacy of 2,4′-BQMs but also elucidated the reaction pathways. This work represents the first successful implementation of organocatalytic enantioselective 1,10-additions to 2,4′-BQMs, establishing a versatile platform for divergent synthesis of two privileged chiral scaffolds from common precursors.

4. Experimental Section

General procedure for products (S)-3: To a solution of $CHCl_3$ (0.3 mL) was added propargylic alcohols **1** (0.06 mmol), 3-phenylindoles **2** (0.05 mmol), and **CPA-2** (0.001 mmol). The reaction mixture was stirred at room temperature for 48 h, and then the solvent was removed under vacuum to give a residue, which was purified by silica gel chromatography to yield products (*S*)-**3**.

General procedure for products (S)-5: To a solution of $CHCl_3$ (0.3 mL) was added propargylic alcohols **1** (0.05 mmol), 2-indole derivatives **4** (0.1 mmol), and **CPA-7** (0.001 mmol). The reaction mixture was stirred at room temperature for 24 h, and then the solvent was removed under vacuum to give a residue, which was purified by silica gel chromatography to yield products (*S*)-**5**.

Acknowledgements

T.L., M.L., and A.U. contributed equally to this work. The authors acknowledge the financial support from the Special Funds of the Taishan Scholar Program of Shandong Province (Grant No. tsqn201812047), the Natural Science Foundation of Shandong Province (Grant No. ZR2021MB026), Guangdong Innovative Program (Grant No. 2019BT02Y335), and the Guangdong Provincial Key Laboratory of Catalysis (Grant No. 2020B121201002). S.-F.N. acknowledge the financial support from the Guangdong Basic and Applied Basic Research Foundation (Grant Nos. 2024A1515010323 and 2025A1515011907), and the open research fund of Songshan Lake Materials Laboratory (grant no. 2023SLABFN16). The authors acknowledge the assistance of SUSTech Core Research Facilities, Xiaoyong Chang (X-ray), Yang Yu (HRMS).

Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the supplementary material of this article.

References

[1] a) S. Zhu, L. Xu, L. Wang, J. Xiao, *Chin. J. Org. Chem.* **2016**, *36*, 1229; b) C. G. S. Lima, F. P. Pauli, D. C. S. Costa, A. S. de Souza, L. S. M. Forezi, V. F. Ferreira, F. de Carvalho da Silva, *Eur. J. Org. Chem.* **2020**, *2020*, 2650; c) X. Li, Z. Li, J. Sun, *Nat. Synth.* **2022**, *1*, 426; d) C. Dorsch, C. Schneider, *Synthesis.* **2022**, *54*, 3125; e) H. Zhang, F. Shi, *Chin. J. Org. Chem.* **2022**, *42*, 3351; f) H. Wang, S. Yang, Y. Zhang, F. Shi, *Chin. J. Org. Chem.* **2023**, *43*, 974; g) Y. Xia, M. Liu, W. Li, P. Li, *Asian J. Org. Chem.* **2024**, *13*, e202400377.

[2] a) T. P. Pathak, M. S. Sigman, *J. Org. Chem.* **2011**, *76*, 9210; b) L. Caruana, M. Fochi, L. Bernardi, *Molecules* **2015**, *20*, 11733; c) Z. Wang, J. Sun, *Synthesis* **2015**, *47*, 3629; d) A. A. Jaworski, K. A. Scheidt, *J. Org. Chem.* **2016**, *81*, 10145; e) W. Li, X. Xu, P. Zhang, P. Li, *Chem. Asian J.* **2018**, *13*, 2350.

[3] a) R. W. Van De Water, T. R. R. Pettus, *Tetrahedron.* **2002**, *58*, 5367; b) N. J. Willis, C. D. Bray, *Chem. Eur. J.* **2012**, *18*, 9160; c) Q. Li, T. Dong, X. Liu, X. Zhang, X. Yang, X. Lei, *Curr. Org. Chem.* **2014**, *18*, 86; d) S. Gnaim, D. Shabat, *Acc. Chem. Res.* **2014**, *47*, 2970; e) M. S. Singh, A. Nagaraju, N. Ananda, S. Chowdhury, *RSC Adv.* **2014**, *4*, 55924.

[4] Z. Yue, F. Fang, Y. Xia, P. Li, W. Li, *Adv. Synth. Catal.* **2023**, *365*, 1926.

[5] S. Liu, K. L. Chan, Z. Lin, J. Sun, *J. Am. Chem. Soc.* **2023**, *145*, 12802.

[6] M. Liu, B. Shen, C. Liu, P. Yu, P. Li, *J. Am. Chem. Soc.* **2023**, *145*, 14562.

[7] a) X. Li, M. Duan, Z. Deng, Q. Shao, M. Chen, G. Zhu, K. N. Houk, J. Sun, *Nat. Catal.* **2020**, *3*, 1010; b) Z. Li, Y. Li, X. Li, M. Wu, M.-L. He, J. Sun, *Chem. Sci.* **2021**, *12*, 11793; c) X. Tan, Z. Deng, Q. Wang, S. Chen, G. Zhu, J. Sun, *Nat. Synth.* **2023**, *2*, 275; d) Z. Han, B. Zhu, Y. Zang, C. Zhang, X.-Q. Dong, H. Huang, J. Sun, *Chem. Sci.* **2024**, *15*, 720.

[8] X. Wang, B. Shen, M. Liu, Z. Wang, C. Liu, P. Li, P. Yu, W. Li, *Angew. Chem., Int. Ed.* **2024**, *63*, e202400143.

[9] a) Y. Shi, P. Wan, *J. Chem. Soc. Chem. Commun.* **1995**, 1217; b) Y. Shi, P. Wan, *Can. J. Chem.* **2005**, *83*, 1306; c) M. Xu, M. Lukeman, P. Wan, *Photochem. Photobiol.* **2006**, *82*, 50; d) N. Basaric, N. Cindro, D. Bobinac, K. Mlinaric-Majerski, L. Uzelac, M. Kralj, P. Wan, *Photochem. Photobiol. Sci.* **2011**, *10*, 1910.

[10] a) L. Zhang, Y. Han, A. Huang, P. Zhang, P. Li, W. Li, *Org. Lett.* **2019**, *21*, 7415; b) M. Liu, C. Qian, P. Li, *Eur. J. Org. Chem.* **2021**, *2021*, e202101415; c) X. Lin, B. Shen, Z. Wang, Y. Cheng, X. Chen, P. Li, P. Yu, W. Li, *Org. Lett.* **2022**, *24*, 4914; d) C. Qian, M. Liu, J. Sun, P. Li, *Org. Chem. Front.* **2022**, *9*, 1234;

 © 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

e) Z. Yue, B. Shen, J. Cao, X. Chen, F. Fang, P. Li, P. Yu, W. Li, *Org. Chem. Front.* **2023**, *10*, 3662.

[11] a) X. Li, M. Duan, P. Yu, K. N. Houk, J. Sun, see a For organocatalytic enantioselective, *Nat. Commun.* **2021**, *12*, 4881; b) Z. Wang, Y. Cheng, Z. Yue, X. Chen, P. Li, W. Li, *Asian J. Org. Chem.* **2022**, *11*, e202200399.

[12] a) J.-F. Bai, L. Zhao, F. Wang, F. Yan, T. Kano, K. Maruoka, Y. Li*Org. Lett.* **2020**, *22*, 5439; b) W.-R. Zhu, Q. Su, X.-Y. Deng, J.-S. Liu, T. Zhong, S.-S. Meng, J.-T. Yi, J. Weng, G. Lu, *Chem. Sci.* **2022**, *13*, 170; c) P. Wu, L. Yu, C.-H. Gao, Q. Cheng, S. Deng, Y. Jiao, W. Tan, F. Shi, *Fundam. Res.* **2023**, *3*, 237; d) S. Pan, X. Wang, M. Liu, F. Fang, Y. Xia, P. Li, W. Li, *ChemCatChem.* **2024**, *16*, e202400900.

[13] a) Q. H. Li, X. Jiang, K. Wu, R. Q. Luo, M. Liang, Z. H. Zhang, Z. Y. Huang, *Curr. Org. Chem.* **2020**, *24*, 694; b) X. Wang, X. Chen, W. Lin, P. Li, W. Li, *Adv. Synth. Catal.* **2022**, *364*, 1212.

[14] a) T. Akiyama, J. Itoh, K. Yokota, K. Fuchibe, *Angew. Chem., Int. Ed.* **2004**, *43*, 1566; b) D. Uraguchi, M. Terada, *J. Am. Chem. Soc.* **2004**, *126*, 5356; c) D. Nakashima, H. Yamamoto, D. Parmar, E. Sugiono, S. Raja, S. M. Rueping, *J. Am. Chem. Soc.* **2006**, *128*, 9626; d) T. Akiyama, K. Mori, *Chem. Rev.* **2015**, *115*, 9277; e) T. James, M. van Gemmeren, B. List, *Chem. Rev.* **2015**, *115*, 9388; f) J. Kikuchi, M. Terada, *Chem. Eur. J.* **2021**, *27*, 10215.

 © 2025 Wiley-VCH GmbH

16154169, 2025, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.70091 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License